How We Trained and Evaluated the Models

Accuracy, Precision, Confusion Matrix — with Real Examples from Our Project

Project: Skin Cancer Detection Using Deep Learning

In one sentence: we trained three models (EfficientNetV2-S, ConvNeXt-T, Swin-T) on the same 18,076 ISIC 2019 training images, combined their answers by averaging, and tested them on 3,596 ISIC 2019 images and 2,298 PAD-UFES-20 phone photos that they had never seen. The combined model classified **2,803 of 3,596** test images correctly (**77.95% accuracy**) with **68.67% precision** (macro average).

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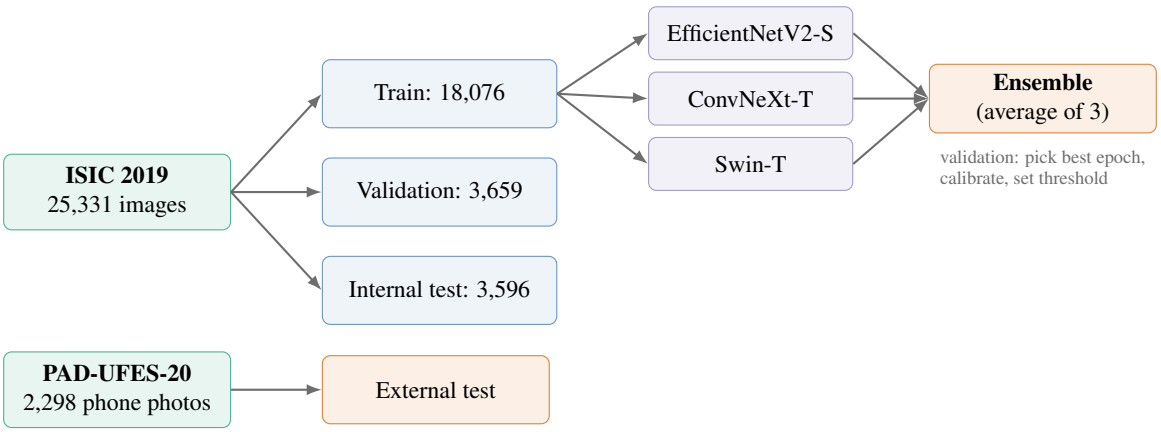
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1 Which Dataset Was Used by Which Model

All three models were trained on **exactly the same data**. This is important: it makes the comparison between the models fair, and it means the ensemble's improvement comes from combining the models, not from extra data.



Model	Trained on	Tuned on	Tested on
EfficientNetV2-S (CNN)	ISIC 2019 train, 18,076	ISIC 2019 validation, 3,659	ISIC 2019 test (3,596) + PAD-UFES-20 (2,298)
ConvNeXt-T (CNN)	same 18,076	same 3,659	same test sets
Swin-T (transformer)	same 18,076	same 3,659	same test sets
Ensemble (ours)	no extra training: averages the 3 models	temperatures, weights, threshold	same test sets

Why this split? Photos of the same mole were always kept in the same part (train, validation, or test), and 344 groups of near-identical photos were merged, so the models are never tested on a mole they have already seen. PAD-UFES-20 was *never* used for training; it checks whether the models work on a different kind of photo.

In the paper: Section III (Datasets), Table II ("Used" rows), Section V-A (Leakage-free protocol).

2 How We Trained the Models

1. **Prepare images.** Every photo was resized (shorter side 256 pixels). During training a random 224×224 crop is taken, flipped, rotated, and colour-changed slightly (*augmentation*), so the model sees a slightly different version every time.
2. **Start from pre-trained models.** Each model had already learned general images (ImageNet-21k). We replaced its last layer with a new one for our 8 skin classes. This is *transfer learning*.
3. **Train with focal loss.** The model looks at a batch of 32 photos, predicts, and the *focal loss* measures how wrong it was (rare classes such as SCC get extra weight). The AdamW optimizer then adjusts the model's weights a little. One pass over all 18,076 photos is one *epoch*.
4. **Check after every epoch.** After each epoch the model is scored on the 3,659 validation photos (balanced accuracy). The best epoch is saved; training stops after 20 epochs or after 6 epochs without improvement (*early stopping*).
5. **Combine.** Each model's confidence is calibrated (*temperature scaling*); then the three probability lists are averaged (*soft voting*).
6. **Test once.** Only at the end are the test sets used.

Model	Epochs run	Best epoch	Best val. BA	Training time (RTX 3060 laptop)
EfficientNetV2-S	20	19	64.16%	43 min
ConvNeXt-T	20	17	64.61%	49 min
Swin-T	11 (early stop)	5	63.05%	36 min

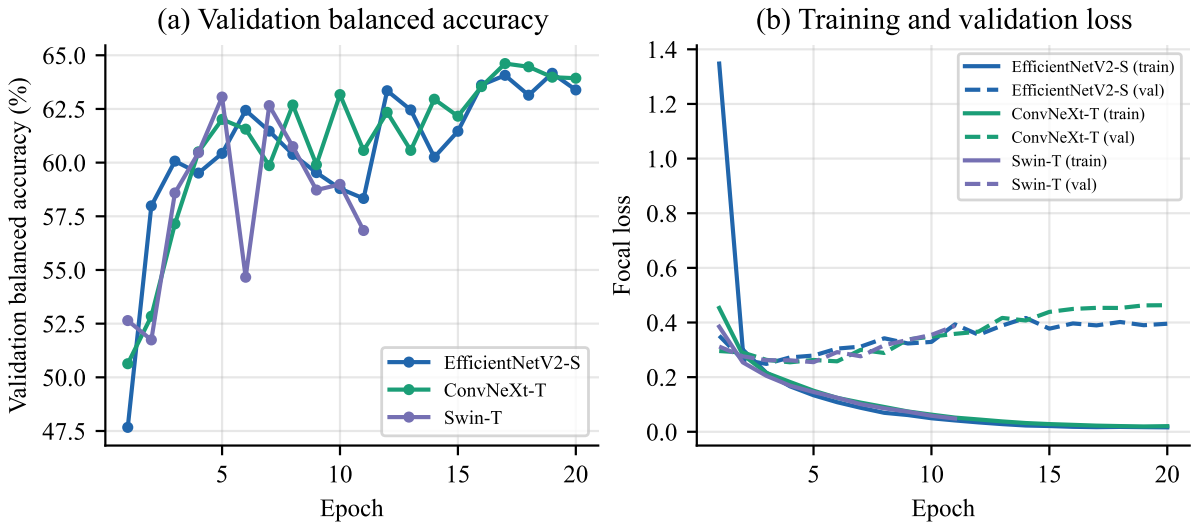
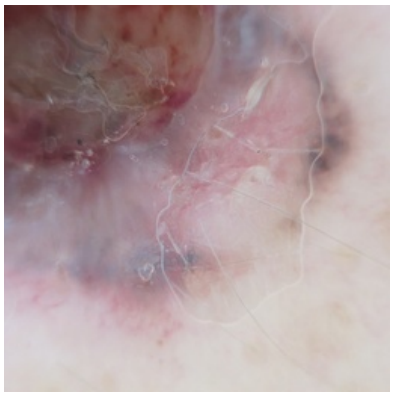


Figure 1: Real training curves. Left: validation balanced accuracy per epoch. Right: loss (solid = training, dashed = validation).

In the paper: Section IV-I (Training procedure), Algorithm 1, Table III (settings), Fig. 4 (training curves), Section VI-A.

3 How One Prediction Is Made (Real Example)



Test image ISIC_0057417
true diagnosis: **melanoma**

Model	MEL	BCC	SCC	Answer
EfficientNetV2-S	27.2%	47.9%	21.4%	BCC (wrong)
ConvNeXt-T	48.2%	49.1%	0.2%	BCC (wrong)
Swin-T	78.4%	13.8%	4.8%	MEL (right)
Ensemble	**51.3%**	36.9%	8.8%	**MEL (right)**

Each model gives a probability for all 8 classes (three shown). The ensemble averages them, e.g. for melanoma $(27.2 + 48.2 + 78.4)/3 = 51.3\%$, and picks the highest. Two models alone were wrong, but the combined answer is right.

In the paper: Equation (4) in Section IV-G (Calibrated ensemble).

4 The Confusion Matrix

A confusion matrix counts, for every true class (rows), what the model predicted (columns). The **diagonal** (top-left to bottom-right) is correct; everything else is a mistake.

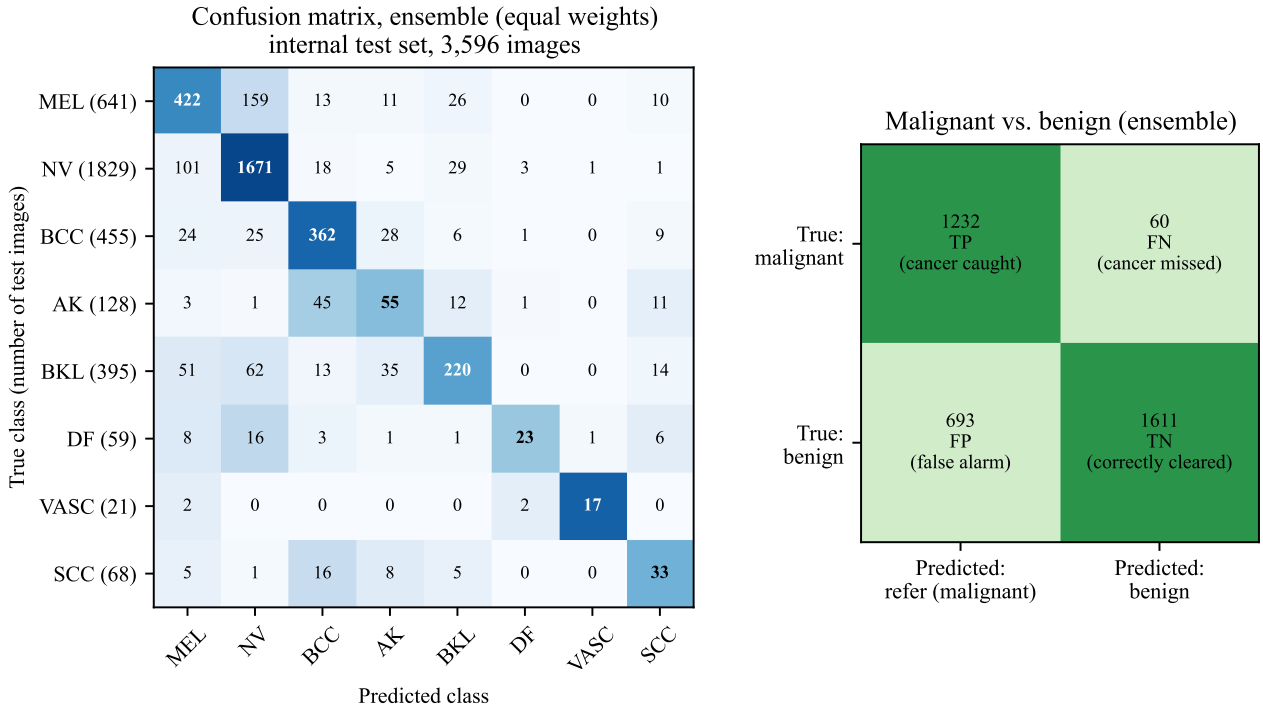


Figure 2: Left: 8-class confusion matrix of the ensemble on the 3,596 internal test images (counts). Right: the same predictions as a 2×2 matrix "malignant (MEL, BCC, SCC, AK) vs. benign" at the referral threshold.

How to read it: the melanoma row says that of the 641 real melanomas, **422** were predicted as melanoma (correct), 159 as nevus, 26 as benign keratosis, and so on. The melanoma *column* says that the model predicted "melanoma" 616 times in total, and 422 of those were real melanomas.

In the paper: Fig. 6 (8-class confusion matrix) in Section VI-C.

5 How Each Metric Is Calculated (with Our Numbers)

For one class: **TP** = true positive (correctly found), **FP** = false positive (wrongly predicted as this class), **FN** = false negative (this class but missed), **TN** = true negative.

Metric	Formula	Worked example from our confusion matrix
Accuracy	$\frac{\text{correct predictions}}{\text{all images}}$	Diagonal sum $= 422+1671+362+55+220+23+17+33 = 2803$; $2803/3596 = \mathbf{77.95}\%$
Precision (MEL)	$\frac{TP}{TP+FP}$ = "when it says melanoma, how often is it right?"	$422/616 = \mathbf{68.5}\%$ (616 = melanoma column)
Recall / sensitivity (MEL)	$\frac{TP}{TP+FN}$ = "of all real melanomas, how many were found?"	$422/641 = \mathbf{65.8}\%$ (641 = melanoma row)
F1-score (MEL)	$\frac{2 \cdot P \cdot R}{P+R}$	$2 \cdot 0.685 \cdot 0.658/(0.685+0.658) = \mathbf{67.1}\%$
Macro precision	average of the 8 class precisions	$(68.5+86.4+77.0+38.5+73.6+76.7+89.5+39.3)/8 = \mathbf{68.7}\%$
Balanced accuracy	average of the 8 class recalls	$(65.8+91.4+79.6+43.0+55.7+39.0+81.0+48.5)/8 = \mathbf{63.0}\%$
Sensitivity (malignant)	$\frac{TP}{TP+FN}$ (2×2 matrix)	$1232/(1232+60) = \mathbf{95.4}\%$ of cancers referred
Specificity (malignant)	$\frac{TN}{TN+FP}$	$1611/(1611+693) = \mathbf{69.9}\%$ of harmless lesions cleared
Precision (malignant)	$\frac{TP}{TP+FP}$	$1232/(1232+693) = \mathbf{64.0}\%$ of referrals are real cancers
AUC	area under the ROC curve (sensitivity vs. $1-$specificity for all thresholds)	0.950 (8 classes, macro); 0.934 (malignant vs. benign). 1.0 = perfect, 0.5 = guessing.

Why are accuracy and balanced accuracy different? Half of the test images are nevi (1,829 of 3,596), and the model is good at nevi (91.4%). Accuracy counts every image equally, so it is pulled up by the nevi. Balanced accuracy gives each of the 8 classes equal weight, so the hard rare classes (AK, SCC, DF) count as much as nevi.

In the paper: Equations (6)–(8) in Section V-B (Metrics).

6 All Results: Accuracy, Precision, Recall, F1

Table 1: Internal test set (ISIC 2019, 3,596 images). Precision, recall, F1 = macro averages.

Model	Correct	Accuracy	Precision	Recall (=BA)	F1	AUC
EfficientNetV2-S	2,704	75.19%	61.66%	58.88%	59.32%	0.943
ConvNeXt-T	2,765	76.89%	67.95%	60.21%	62.68%	0.936
Swin-T	2,628	73.08%	57.84%	57.80%	56.47%	0.935
Ensemble (equal)	**2,803**	**77.95%**	**68.67%**	**62.99%**	**64.79%**	**0.950**
Ensemble (learned)	2,765	76.89%	65.82%	61.65%	62.47%	0.950

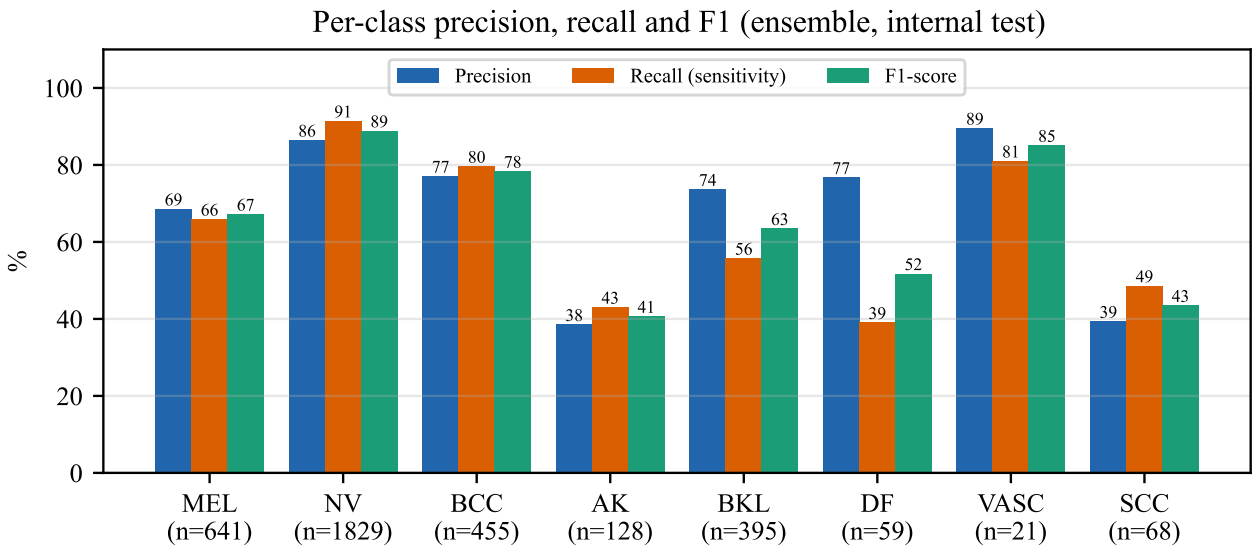


Figure 3: Precision, recall, and F1 for each class (ensemble, internal test).

Table 2: External test set (PAD-UFES-20, 2,298 phone photos, 6 classes).

Model	Accuracy	Precision	Recall (=BA)	F1	Cancer AUC
EfficientNetV2-S	33.68%	34.75%	39.74%	28.50%	0.833
ConvNeXt-T	34.73%	33.82%	38.22%	28.64%	0.844
Swin-T	35.34%	35.34%	41.50%	30.03%	0.844
Ensemble (equal)	**37.25%**	**37.05%**	**41.89%**	**30.74%**	**0.873**

In the paper: Table IV (internal, with precision), Table V (per-class precision/recall/F1), Table VII (external), Fig. 5 (bar charts), Fig. 7 (ROC), Table VI (ablation: 1 vs. 2 vs. 3 models).

7 Example Images from the Test Set

(a) Correct predictions — one per class

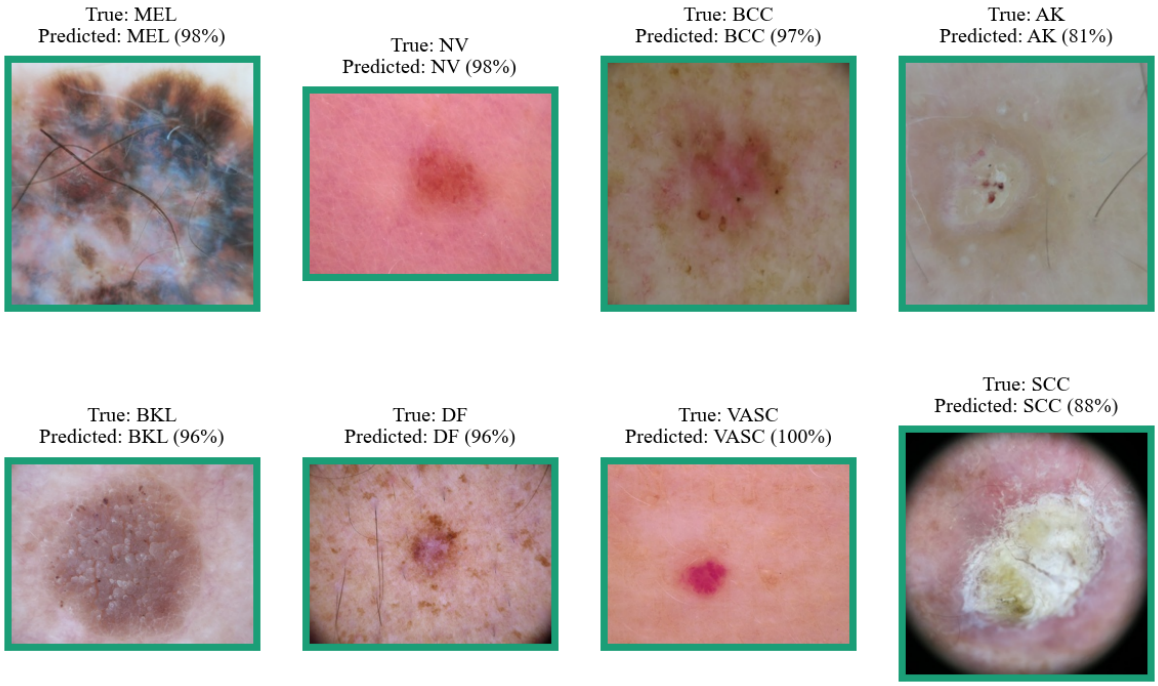


Figure 4: Real test images (never seen in training). Green frame = correct. The percentage is the ensemble's confidence.

(b) Why combining models helps (our USP)

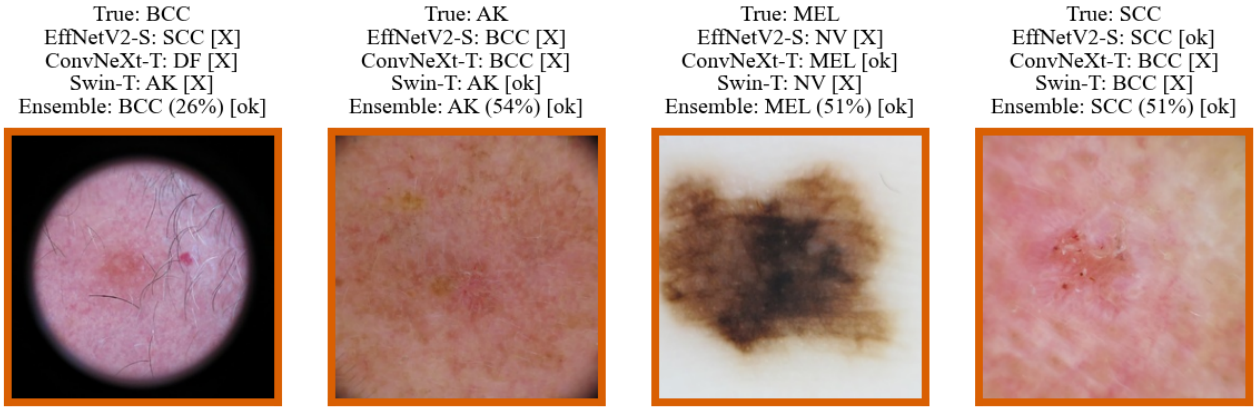


Figure 5: Cases where one or more single models were wrong but the ensemble was right. In the first image all three models were wrong individually, yet the average of their probabilities picked the correct class. In the internal test set this happened 556 times; the opposite (ensemble wrong although at least one model was right) happened 290 times.

(c) Typical mistakes

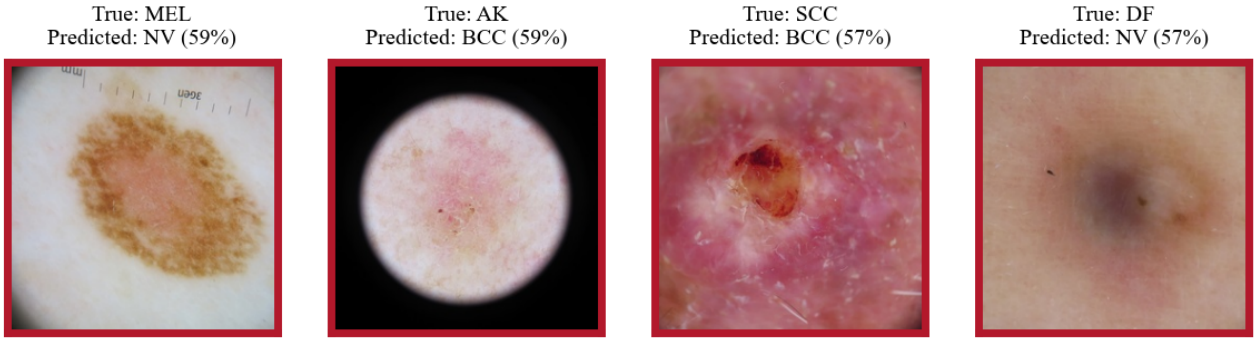


Figure 6: Red frame = wrong. These are the most common confusions in the confusion matrix: melanoma vs. nevus, AK/SCC vs. BCC, dermatofibroma vs. nevus. They are also hard for doctors.

(d) Phone photos (PAD-UFES-20)

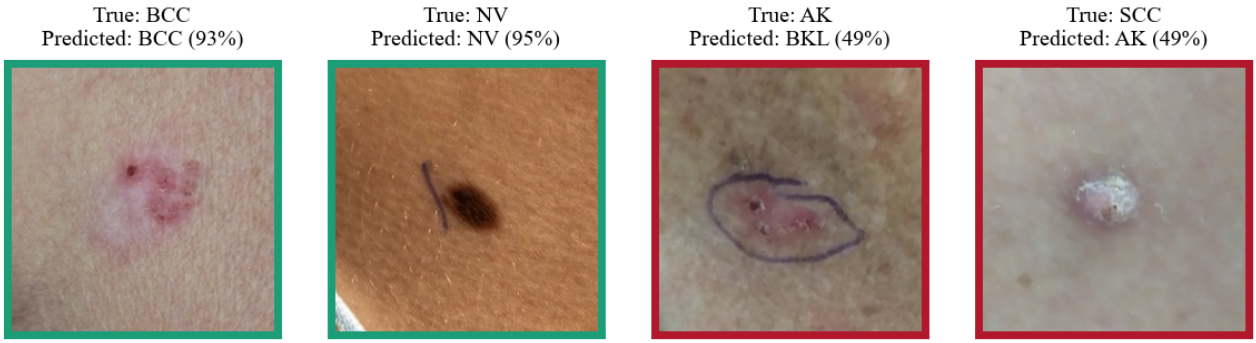


Figure 7: External test. Phone photos look different from dermoscope close-ups, which is why accuracy drops to 37%.

8 Where Everything Is in the Paper

Item	Location in `sc_two_column.pdf`
Datasets used and split sizes	Section III; Table II (rows marked "Used"); Section V-A
Which models, pre-trained weights	Section IV-D (Hybrid backbone ensemble)
How we trained (steps, settings, time)	Section IV-F (focal loss, Eq. 3), IV-I (training procedure), Algorithm 1, Table III
How models are combined	Section IV-G, Eq. (4)
How accuracy, precision, recall, F1 are calculated	Section V-B, Eqs. (6)–(8)
Training graph	Fig. 4 (Section VI-A)
Accuracy and precision of every model	Table IV and Fig. 5(a) (Section VI-B)
Confusion matrix	Fig. 6 (Section VI-C)
Per-class precision / recall / F1	Table V (Section VI-C)
ROC curve	Fig. 7 (Section VI-C)
USP proof (1 vs. 2 vs. 3 models)	Table VI (Section VI-D)
External phone-photo results	Table VII and Fig. 5(b) (Section VI-E)
Skin-type results	Section VI-F
Final accuracy and conclusion	Abstract and Section IX

What to tell your teacher (30 seconds): "We trained three models — two CNNs and one transformer — on the same 18,076 ISIC 2019 images, chose the best epoch on 3,659 validation images, and averaged their predictions. On 3,596 unseen test images the combined model was right 2,803 times, which is 77.95% accuracy, with 68.67% precision, 62.99% balanced accuracy, and an AUC of 0.950. It found 95.4% of the cancers. The confusion matrix in Fig. 6 shows where it makes mistakes, mainly melanoma versus nevus. Combining the models beat every single model and every pair, which is our USP. On phone photos it drops to 37%, which shows what we need to improve next."